

No column above the one before it in every row: OEIS A212941

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Abstract

OEIS A212941 carries an empirical recurrence of order 11 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition quantifies over all rows at once, so it is not decided by any bounded window; but it is a conjunction of independent per-row facts, and one bit per adjacent column pair carries it; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 11$ vertices and is C -finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A212941 is “Number of $n \times 6$ 0..5 arrays with no column j greater than or equal to than column $j-1$ in all rows.”. It has offset 1 and begins

1, 152747323, 28744943858947, 2438624218076957695, 153322267564381742818531, 8389667096561737605103329

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 130453*a(n-1) - 6763333836*a(n-2) + 185208872760324*a(n-3) - 2991440828618168976*a(n-4) + 29966228467174160130816*a(n-5) - 189618846625740940674481152*a(n-6) + 754213725094693350993932353536*a(n-7) - 1831929539263722316502211827073024*a(n-8) + 2545588680910314328406052486644760576*a(n-9) - 1758206800538810904218951296840095498240*a(n-10) + 418951577236630476027552063815969641857024*a(n-11)$.

As of the “Last modified” line on the live entry (Apr 17 2022 21:53:33, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays of 6 columns whose rows are arbitrary over the alphabet $\{0, 1, \dots, 5\}$, and asks that no column j be greater than or equal to column $j - 1$ *in all rows*.

That is not a condition on any bounded window: “in all rows” quantifies over the whole array, and a row arriving at the very end can still decide it. What makes it a walk anyway is that it is a conjunction of independent facts about single rows. For each of the 5 adjacent column pairs keep one bit, recording whether every row so far has had column j greater than or equal to column $j - 1$. A new row can only clear bits, never set one, and the array is admissible exactly when every bit has been cleared by the time the array ends.

So the state is that bit mask and nothing else — at most 2^5 of them, however large the alphabet is, which is why entries whose terms run to twenty digits have models this small. Rows

with the same bit pattern act identically, so each row contributes one edge and the multiplicity does the counting. Merging vertices with identical futures leaves $S = 11$, and with M the adjacency matrix and ι, τ the vectors recording which masks may begin and end an array,

$$a(n) = \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^{11} - \sum_i c_i t^{11-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+11} - \sum_i c_i M^{j+11-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. $\square$

Theorem 2.

$$a(n) = 130453 a(n-1) - 6763333836 a(n-2) + 185208872760324 a(n-3) - 2991440828618168976 a(n-4) + 2996661111111111111 a(n-5) - 130453 a(n-6)$$

for every $n > 11$.

Proof. The vector $w = q(M) \tau$ was formed by 11 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 11 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 8 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one and tests, for each adjacent column pair, whether every row really does put one above the other. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.